



Negasi Teklay Weldesemat

Phone: (+82) 1021501087 (Mobile) | Email address: negasi2020@mju.ac.kr, negasi@dgist.ac.kr | Address: Daegu, South Korea (Work)

Work experience

Research Assistant | Department of Energy Science & Technology, Myongji University | 01/09/2020 - 03/01/2025 | Yongin, South Korea

Hand-on experience on synthesis of organic compounds (crown ether derivatives), fabrication of conducting polymer and layered cathode material, coin cell assembly, development of smart materials such as electro -, photo -, and pH -responsive materials.

Worked with advanced equipment such as electrospinning, scanning electron microscopy, FTIR, UV-Vis, BET, electrochemical workstations, ICP-MS, chromatography, muffle furnace, etc.

Teaching Assistant | Myongji University | 15/02/2021 - 30/06/2021 | Yongin-si, South Korea

Presented lecture on fundamental principle and instrumentation of GC-MS, HPLS, FTIR, NMR, XRD, XPS, DSC-TGA, Raman Spectroscopy and ICP-MS;

Trained student in FTIR, XRD and ICP-MS.

Chemical Engineering Intern | Almeda Textile PLC (ALTEX) | 01/09/2017 - 31/01/2018 | Adwa, Ethiopia

Acquired knowledge in process control loops and understanding of process variables and analog/digital signals.

Worked on diverse electrical and electronic device utilized in industrial process control systems, fostering a comprehensive understanding of how various materials and system integrate into transmitters, receivers, and controllers.

Gained broader aspects of organization management and workplace culture, industrial safety procedures and maintenance protocols.

Education & Training

PhD in Energy Science and Engineering | DGIST | 25/08/2025 - Current | Daegu, South Korea

Enrolled courses: Advanced Quantum mechanics, Soft Material and Devices, Mathematics for Machine Learning and Thermodynamics of Energy Systems

Thesis: Developing Machine Learning and High-Throughput Approaches for the Design and Synthesis of N-Type Redox-Active Molecules for Semiconducting Devices

MSc. In Energy Science and Technology | Myongji University | 01/08/2025 | Yongin-si, South Korea

Final grade: 4.38/4.5 | **Thesis:** Development of Electroactive Composite Materials toward Electrochemical Extraction and Recovery Lithium and Silver Ions | **Address:** Myongji-ro 116, Cheoin-gu

BSc. In Chemical Engineering | Mekelle Institute of Technology (Mekelle University) | 01/07/2019 | Mekelle, Ethiopia

Final grade: 3.93/4.00 | **Address:** Aynalem1632

Machine Learning Specialization | Coursera | 07/12/2022 - 03/08/2023

Link: <https://www.coursera.org/account/accomplishments/certificate/XDJHL8QMSQ8U>, <https://www.coursera.org/account/accomplishments/certificate/6BTDSTYD4X6V>

Density Functional Theory | Coursera | 23/01/2023 - 28/07/2023

Link: <https://www.coursera.org/account/accomplishments/certificate/YTGCVVLF838E>

Python for Data Science& AI | Coursera | 07/12/2022 - 05/01/2023

Link: <https://www.coursera.org/account/accomplishments/certificate/ZT5EP4RBU6DW>

Skills

C Programming | Simulink | Python | Principal Component Analysis | GNN | ChemDraw | Design-Expert | Project management tools (Monday.com) | MATLAB | Adobe Illustrator | EdrawMax | Semiconductor laser cutting and 3D printing | Fabrication & Performance Evaluation of RRAM

Publications

Thiolated Hyper-Crosslinked Tetraphenylborate with Tunable Selectivity for Sequential Recovery of Pd and Pt from Autocatalyst Leachate

2024.

Knitted" tetrabenzo-24-crown-8 ether sulfonate as reusable and practical hydrophilic adsorbent for aqueous Cs⁺ sequestration

2024.

<https://doi.org/10.1016/j.jece.2023.111690>

Carbonyl-interfaced-biochar derived from unique capillary structures via one-step carbonization with selective methyl blue adsorption capability

2023.

<https://doi.org/10.1016/j.jclepro.2023.137291>

A Smart Composite Materials (TCE@GO@PPy) based on Thia-crown Ether and Polypyrrole for Electrochemical-assisted Ag⁺ Removal from Wastewater

(Negasi T. Weldesemat, Anteneh F. Baye, Hern Kim), Ready for submission, Target Journal: ACS Applied Materials & Interfaces

Exploring of Aqueous stable Low-Ni Li (NixMnyCoz)O₂ for Optimal Lithium extraction performance: comparison of the structural and electrochemical properties

(Negasi T. Weldesemat, Anteneh F. Baye, Hern Kim), Manuscript Preparation, Target Journal: Journal of Power Sources

Conferences & Seminars

Fine-Tuning the Li⁺ Insertion Capacity and Cyclic Stability of LiNi_{1-x}yMnxCoyO₂ in Aqueous Environments

(Negasi T. Weldesemat, Grace M. Nisola, Hern Kim and Wook-Jin Chung), 2024 Spring Meeting of the Korean Electrochemical Society (KECS), 3-5 April 2024, BEXCO, Busan, South Korea.

Elevating Lithium-Ion Recovery: Breakthroughs in Modified Nickel-Rich Layered Cathodes

(Negasi T. Weldesemat, Grace M. Nisola, Hern Kim and Wook-Jin Chung), 2023 Fall Meeting of the Korean Electrochemical Society, 1-3 November 2023, Yeosu Expo Convention Center, Yeosu, South Korea

A Smart Electrochemical Material for Recovering Ag⁺ from Wastewater

(Negasi T. Weldesemat, Grace M. Nisola, Hern Kim and Wook-Jin Chung), Fall 2023 Korean Institute of Chemical Engineers (KICHE) General Assembly & International Conference, 25-27 October 2023, Daejeon Convention Center, Daejeon, South Korea

Electrochemical Recovery of palladium (II) ion from Acidic leachate

(Negasi T. Weldesemat, Grace M. Nisola, Tilahun K. Mathewos and Wook-Jin Chung) 2021 Korean Society of Environmental Engineers (KSEE) National Conference, 03-05 November 2021, Jeju Shinhwa World, Jeju, South Korea.

Magnetic Graphene Oxide Grafted with Temperature-Responsive Crown Ether Polymer Brushes as an Adsorbent for Lithium Recovery from Seawater

(Khino Parohinog, Negasi T. Weldesemat, John Edward L. Sio, Grace Nisola, Wook-Jin Chung), 2022 AIChE Annual Meeting, 13-18 November 2022, Phoenix Convention Center, Phoenix, AZ, USA;
<https://www.aiche.org/academy/conferences/aiche-annual-meeting/2022/proceeding/paper/171g-magnetic-graphene-oxide-grafted-temperature-responsive-crown-ether-polymer-brushes-adsorbent>

Co_{1-x}[Fe(III)(CN)₆]/Ag for electrochemical recovery of cobalt ion from spent nickel metal hydride batteries

(Hana G. Zeweldi, Gebremedhn T. Gebremichael, Negasi T. Weldesemat, Grace M. Nisola, Wook Jin Chung) 2022 AIChE Annual Meeting, 13-18 November 2022, Phoenix Convention Center, Phoenix, AZ;
<https://www.aiche.org/academy/conferences/aiche-annual-meeting/2022/proceeding/paper/372g-co1-xfeiiicn6ag-electrochemical-recovery-cobalt-ion-spent-nickel-metal-hydride-batteries>

Effective Neodymium (III) recovery from industrial waste by phosphorus-doped mesoporous carbon

(Gebremedhn T. Gebremichael, Negasi T. Weldesemat, Grace M. Nisola and Wook-Jin Chung), 2024 Korean Society of Environmental Engineers (KSEE) national Conference, 09-11 November 2022, Jeju Shinhwa World, Jeju, South Korea.

Selective recovery of cobalt ion from spent from spent nickel metal hydride batteries via electrochemistry

(Hana G. Zeweldi, Tilahun K. Mathewos, Negasi T. Weldesemat, Grace M. Nisola, Wook Jin Chung), 2022 Korean Society of Chemical Engineers Spring General Meeting & Conference, 20-23 April 2022, ICC, Jeju, South Korea

Selective and sustainable recovery of Au³⁺ through complexation-reduction capture and mechano-assisted release by thermo-responsive poly(NIPAM-co-15TCE-4)@SiO₂ nanoparticles

(Hiluf T. Fissaha, Grace M. Nisola, Tilahun K. Mathewos, Negasi T. Weldesemat, Wook-Jin Chung), 2022 Korean Society of Chemical Engineers Spring General Meeting & Conference, 20-23 April 2022, ICC, Jeju, South Korea

Selective recovery of lithium ion from aqueous sources via a three-electrode electrochemical system

Projects

Development of efficient hyper-crosslinked adsorbents based on the ligand with a high affinity toward specific metal using the ligand-metal complexation mechanism | 01/06/2023 - 03/01/2025

Role: Graduate Student Researcher; Funding: NRF of Korea

Development of Sequential and eco-friendly separation process without secondary waste generation using thermos-and-electroresponsive molecules to produce valuable metals | 01/09/2021 - 03/01/2025

Role: Graduate Student Researcher; Funding: NRF of Korea (No. 2021R1A2C2093746)

Development of eco-friendly process for the separation of radioactive Cs⁺ adsorbent with photo-responsive and ion-recognition molecules | 01/03/2021 - 28/02/2023

Role: Graduate Student Researcher; Funding: NRF of Korea (No. 2021R1H1A200828411)

Development of thin-film composite FO membrane with 2-D nanosheets in seawater desalination | 01/06/2022 - 31/05/2023

Role: Graduate Student Researcher; Funding: NRF of Korea

Development of novel draw solute in FO membrane system using DES-hydrogel | 01/01/2022 - 31/05/2022

Role: Graduate Student Researcher; Funding: NRF of Korea

Development of tailor-made hybrid smart materials for the selective separation of precious metals from the electronic wastes | 01/09/2020 - 01/02/2021

Role: Graduate Student Researcher; Funding: NRF of Korea (No. 2017R1A2B2002109)

Networks and Memberships

Korean Electrochemical Society | 01/01/2021 - Current

Korean Society of Environmental Engineers | 01/01/2021 - Current

Korean Institute of Chemical Engineers | 01/01/2021 - Current

Institute of Chemical Engineers | 01/01/2021 - Current

Driving Licence

Driving licence B | 05/08/2022 - 01/01/2032

Recommendations

MSc Supervisor | Hern Kim | (+82) 1043237652 | hernkim@mju.ac.kr

Professor, Department of Energy Science and Technology, Myongji University (South Korea)

BSc. Supervisor | Solomon Kahsay Gebremariam | (+971) 562575382 | solomon.gebremariam@ku.ac.ae

PhD Candidate, Department of Chemical and Petroleum Engineering, Khalifa University, Abu Dhabi, 127788 United Arab Emirates.

Former Lecturer and Department Head, Mekelle University, Mekelle, Ethiopia

I hereby declare that the information provided above are true, complete, and accurate to the best of my knowledge.

